DATA MANAGEMENT CARPENTRY · EPISODE 8

Best Practices for Tabular Data

Core principles of tidy data, clean header architecture, spreadsheet cleaning techniques, and version control for datasets.

EPISODE SCOPE

Questions & Objectives

Core Questions

What makes a tabular dataset "bad"?

How do inconsistent headers cause research errors?

How can standard spreadsheet tools clean messy data?

How should we track versions of data files?

Learning Objectives

Identify pitfalls (stacked tables, missing units).

Explain principles of tidy rectangular data.

Apply Find & Replace for bulk standardization.

Evaluate data version control strategies.

CONTEXT

The Problem: Ubiquitous but Messy

Designed for Eyes, Not Scripts

Spreadsheets like Excel are built to be visually flexible for human readers. This encourages formatting habits that make automated parsing nearly impossible:

- Columns with shifting names for identical metrics
- Units omitted or hidden in hover comments
- Blank separator rows splitting sub-tables

The Downstream Impact

When messy tables are fed into R, Python, or SPSS, scripts silently fail or produce corrupted visualizations.

Clean structure must precede statistical analysis.

DISCUSSION 1 · 5 MINUTES

Spotting Pitfalls in `Incidence-Data(bad).csv`

RAW DATA SNIPPET

```
ID,table,diagnosis Year,Age Specific Incidence Age0_49,age Specific Incidence  Age 0-54 ...
1,All Oral Cavity,All Years,,,,,,,,,
2,All Oral Cavity,2016,1.02,7.15,11.54,14.97 ...
3,All Oral Cavity,2017,1.10,7.77,10.90,13.12 ...
4,All Oral Cavity,2018,0.97,7.10,10.33,13.69 ...
5,All Oral,,0.89,6.51,9.90,12.55,18.08 ...
6,All Oral,2020,0.80,6.84,9.96,12.89,15.29 ...
```

Inspect in Pairs:

- What is wrong with the column header names?
- How are categories labeled down column B?
- What happens with blank cells in row 5?
- Where are the units of measurement?

THE CORE CONFLICT

Human-Readable vs. Machine-Readable

Human Eyes (Forgiving)

Humans instantly deduce that `All Oral` and `All Oral Cavity` represent the same group.

A blank cell beneath "2018" is intuitively assumed to be "2019" carrying over.

Computers (Strictly Literal)

`All Oral` and `All Oral Cavity` are treated as two distinct, unrelated categories.

A blank year is treated as `NaN`, silently dropping rows from statistical models and plots.

DIAGNOSTIC AUDIT

Five Common Pitfalls in the Dataset

1. Inconsistent Header Conventions

Double spaces, camelCase mixed with snake_case, and typos (`age  Spes Incidence`).

2. Missing Units in Headers

Are metrics per 1,000, per 100,000, or raw counts? Numbers are completely ambiguous.

3. Stacked Sub-Tables

Blank divider rows and summary rows (`All Years`) mixed directly into annual data.

4. Inconsistent Categories

Label changes from `All Oral Cavity` to `All Oral` halfway down column B.

5. Implicit Missing Data: Leaving cells blank assuming reader knows values carry over from above.

ARCHITECTURE

Core Principles of Tidy Tabular Data

1. Column = Variable

Each variable forms its own column. Never mix metrics or drop prefixes across headers.

2. Row = Observation

Each row represents a single discrete entity, timepoint, subject, or experimental run.

3. Cell = Single Value

Never combine numbers and text notes, units, or multiple measurements in one cell.

STANDARDIZATION

Golden Rules for Clean Headers

Rules of Thumb

- Use lowercase letters throughout
- Replace spaces with underscores (_)
- Avoid special symbols: ? $ % - / #
- Include units explicitly in the header name

TRANSFORMATION EXAMPLES

Poor Header:

```
Age Specific Incidence Age 0-54
```

Clean Header:

```
incidence_per_100k_age_0_54
```

PRACTICAL WALKTHROUGH

Demonstration: Cleaning Data in Excel

Step 1: Standardise Headers

Select Row 1. Replace spaces with underscores, fix typos (`Spes`), and add units (`per_100k`).

Step 2: Find & Replace

Press `Ctrl + H` (or `Cmd + H`). Find `All Oral` and replace with `All Oral Cavity` across column B.

Step 3: Un-Stack & Fill

Populate missing years explicitly. Delete blank divider rows (10 & 20). Separate summary totals.

HANDS-ON EXERCISE

Challenge: Cleaning the Incidence Dataset

Working on your copy of `Incidence-Data(bad).csv`:

1. Standardise inconsistent category names in the `table` column.
2. Fix headers for `Age85_89` and `Age 90` to match the standardized format.
3. Fill in missing values in the `diagnosis_year` column explicitly.
4. Delete all blank divider rows and separate summary statistics.
5. Save the modified file as `Incidence-Data_clean.csv`.

CHALLENGE SOLUTION

A Tidy, Standardised Dataset

Key Clean Attributes

- Uniform, lowercase snake_case headers across all columns
- Measurement unit (`per_100k`) clearly embedded in headers
- Zero blank divider rows disrupting parsing scripts
- Explicit year values populated in every single row
- Single consistent category: `All Oral Cavity`

Preserve Raw Data

Golden Rule: Never overwrite the raw original file. Save cleaned results as a separate file (e.g. `data_clean.csv`).

Raw data represents your baseline provenance; cleaned data represents analysis input.

DATA EVOLUTION

Version Control for Tabular Data

The Manual File-Naming Trap

```
Incidence-Data_clean.csv
Incidence-Data_clean_v2.csv
Incidence-Data_clean_v2_final.csv
Incidence-Data_clean_v2_final_FINAL_corrected.csv
```

Quickly obscures which file is current, and fails to track what changed, why, or by whom.

Strategic Principles

- **Separate Folders:** `data/raw/` vs `data/processed/`
- **Cloud Revision History:** OneDrive / SharePoint snapshots
- **Formal Git:** Line-by-line diffs for text CSVs and scripts

SUMMARY

Episode 8 Key Points

Machine-Readability: Layout tables strictly for automated scripts, not human visual preference.

Tidy Rules: One variable per column, one observation per row, and one value per cell.

Clean Headers: Use lowercase letters, underscores instead of spaces, and state units explicitly.

Protect Raw Data: Keep raw files read-only and track changes with cloud history or Git.

The Invariant: Clean tabular structure ensures your research is automated, reproducible, and verifiable.

LOOKING AHEAD

Next: Episode 9

Now that we have structured our tabular spreadsheets into clean rectangular data, we explore choosing appropriate, sustainable file formats for long-term preservation.

Choosing Open & Preservable File Formats